

Who Pays for the Oscillation?

Exact Shapley Attribution of Dynamic-Security Interactions via Prime Intervention Cycles in Interconnection Cluster Studies

Jorge L. Mayorga Taborda

Universidad de los Andes, Bogotá, Colombia

Manuscript draft, August 2026

Abstract

Interconnection queues are studied in clusters: dozens of projects with different owners are evaluated together, and the cost of the network upgrades and mitigations that the cluster triggers must be allocated among them. For *static* quantities (flows, losses, thermal upgrades) mature allocation methods exist. For *dynamic-security* interactions—the converter-driven oscillations behind the Odessa and Hornsea events, and the mechanism-level IEEE 39-bus counterexample in which two individually beneficial projects jointly destroy 10.76% of the damping margin—there is no causal allocation rule at all: when a jointly created oscillatory mode forces a retrofit, no existing method says whose mode it is. This paper provides one, and proves that it is exactly computable. Modeling a cluster of m candidate actions as a finite-rank perturbation of a controller-aware grid operator, the portfolio spectral shift $\Phi(S) = \log \det(I + K_S)$ defines a transferable-utility cooperative game on the actions whose interaction matrix restricts to principal submatrices. Our main theorem shows that the Harsanyi dividend of every coalition R equals the aggregate log-Euler factor of the *prime intervention cycles supported exactly on R* ; pairwise dividends collapse to a single closed-form factor $\log(1 - q_{ab}q_{ba})$. Consequently the Shapley value of each project—its causally defensible share of the cluster’s dynamic externality—has a closed form: its isolated term plus an equal share of every prime cycle it participates in, computable to certified tolerance in $O(m^p)$ by Lyndon-word enumeration, although Shapley values of general succinct games are #P-hard. Three further results make the rule usable in regulation and design. *Cluster-composition independence*: a coalition’s assessed synergy does not change when unrelated projects join or leave the queue, closing a gaming channel that pro-rata rules leave open. *Mode-count attribution*: on any frequency contour where the interaction radius is certified, the integer number of portfolio-created modes decomposes exactly into winding numbers of finitely many identifiable

prime cycles—“whose mode it is” becomes a theorem, not a negotiation. *Cycle surgery*: because the certified radius $\rho(|Q|)$ is monotone and Perron-differentiable in the cycle-edge magnitudes, the cheapest retuning that breaks a destructive cycle is found by a Perron-sensitivity targeting rule, formalizing the observed single-PLL repair of the IEEE 39-bus stress case. On the flagship destructive portfolio, the rule allocates the joint margin loss 68.3% to the line project and 31.7% to the retuning, while certifying that any project outside the feedback core owes nothing beyond its own term—the but-for causation standard, derived rather than asserted.

1 Introduction

1.1 The allocation gap

Modern interconnection processes evaluate generation and storage projects in clusters. Under cluster-study reforms such as FERC Order No. 2023 [12], the projects in a queue window are studied together and the cost of triggered network upgrades is allocated among them. Allocation practice, and its supporting literature, is built for *static* quantities: MW-mile and usage-based methods, flow tracing [11], and cooperative-game allocations of losses and transmission expansion cost [10]. All of these divide something that superposes.

Dynamic security does not superpose. In inverter-dominated grids, the small-signal consequence of a set of projects and mitigations is governed by feedback cycles that the actions close *through each other* via the network-controller resolvent. A mechanism-level IEEE 39-bus campaign [2] produced the canonical counterexample: a line reinforcement worth +1.45% of damping margin alone and a controller retuning worth +5.39% alone jointly yield −10.76%. Field events show the same physics at commercial scale: converter-driven oscillations at Odessa [13] and the 9 August 2019 Hornsea event in Great Britain [14] triggered retrofits, curtailment, and—centrally for this paper—*disputes about responsibility* that were settled

by negotiation, because no method existed to compute whose oscillation it was.

This paper supplies the missing object: an exact, axiomatically grounded, computationally tractable attribution of dynamic-security interactions to the projects that cause them.

1.2 Approach and contributions

The companion theory [1, 2, 4] represents a portfolio $\mathcal{A} = \{a_1, \dots, a_m\}$ of localized actions as a finite-rank perturbation of a controller-aware operator $T_0(s)$, encodes its entire spectral consequence in an $m \times m$ action Green matrix $K_{\mathcal{A}}(s)$ with $\det T_{\mathcal{A}} / \det T_0 = \det(I + K_{\mathcal{A}})$, and proves that the normalized interaction correction factorizes as an Euler product over *prime intervention cycles*—aperiodic closed feedback loops in action space—in the sense of the Bowen–Lanford and Ihara–Stark–Terras zeta functions [15, 16, 17, 18]. The present paper turns that factorization into an economics of responsibility. Contributions:

- C1. The portfolio spectral game.** The set function $\Phi(S) = \log \det(I + K_S(s))$ on subsets of $\mathcal{A}$ is a well-defined transferable-utility game whose interaction data restrict by principal submatrices (Lemma 1).
- C2. Dividend–cycle correspondence (Theorem 1).** The Harsanyi dividend of every coalition R with $|R| \geq 2$ equals the sum of log-Euler factors of the prime cycles supported *exactly* on R . Pair dividends are single closed-form factors; the Möbius interactions measured empirically in [2] acquire an exact algebraic identity.
- C3. Closed-form Shapley attribution (Theorem 2).** Each action’s Shapley share of the cluster’s spectral shift is its isolated term plus an equal split of every prime cycle it belongs to—computable to certified tolerance in $O(m^p)$ by Lyndon-word enumeration, versus the #P-hardness of Shapley values for general succinctly represented games [8].
- C4. Cluster-composition independence (Theorem 3).** Coalition dividends do not depend on which other projects are in the study snapshot, so the allocation cannot be gamed by queue composition—a property pro-rata rules provably lack.
- C5. Mode-count attribution (Theorem 4).** On any frequency contour with certified interaction radius, the integer count of portfolio-created modes splits exactly into winding numbers of finitely many identifiable prime cycles: “whose unstable mode” is decided by an integer-valued theorem.
- C6. Cycle surgery (Section 6).** Monotonicity and Peron differentiability of the certified radius $\rho(|Q|)$ yield a targeting rule for the cheapest retuning that dissolves a destructive feedback core, formalizing the single-PLL repair observed in the IEEE 39-bus stress case [3].

Everything game-theoretic used here—Shapley values [5], Harsanyi dividends [6], Möbius interaction indices [7]—is classical, as is the zeta machinery [15, 16, 17]. What is claimed as new is the identification of the dividends of a *dynamic-stability* functional with prime-cycle bundles, the exactness and tractability that follow, and the resulting allocation and design rules. Cooperative-game allocation in power systems has, to date, addressed static cost and loss sharing [10, 11]; attribution of small-signal interactions appears to be unaddressed. Shapley-based attribution of black-box models (SHAP [9]) is the machine-learning analogue of what is done here in closed form for a physical determinant.

2 Preliminaries

2.1 Controller-aware operator and action Green matrix

After Schur elimination of controller states, the retained grid dynamics are $T_0(s) = s^2 M + s \Sigma(s) + L$ with a frequency-dependent, non-diagonal self-energy $\Sigma(s)$ [2]. A portfolio $\mathcal{A} = \{a_1, \dots, a_m\}$ of elementary actions (line, inertia, damping, BESS/GFM, controller retuning) enters as $T_{\mathcal{A}} = T_0 + U_{\mathcal{A}} C_{\mathcal{A}} V_{\mathcal{A}}^*$, and with $G_0 = T_0^{-1}$,

$$K_{\mathcal{A}}(s) = C_{\mathcal{A}} V_{\mathcal{A}}^* G_0(s) U_{\mathcal{A}}, \quad \frac{\det T_{\mathcal{A}}}{\det T_0} = \det(I + K_{\mathcal{A}}). \quad (1)$$

Split $K = D_K + E_K$, $D_K = \text{diag}(K)$, and (for $1 + K_{aa} \neq 0$) define the zero-diagonal normalized interaction matrix $Q = (I + D_K)^{-1} E_K$, so that $\det(I + K) = \prod_a (1 + K_{aa}) \det(I + Q)$.

2.2 Prime cycles and the Euler product

A closed walk $\gamma = (a_1 \rightarrow \dots \rightarrow a_\ell \rightarrow a_1)$ in the action digraph carries weight $q(\gamma) = q_{a_1 a_2} \dots q_{a_\ell a_1}$; cycles are walks modulo rotation, and a cycle is *prime* if not a repetition of a shorter one. Let $\mathcal{P}$ be the set of prime cycles and $\text{supp}(\gamma) \subseteq \mathcal{A}$ the set of distinct actions γ visits. The companion result [1] is: for $\rho(|Q(s)|) < 1$ (entrywise absolute value),

$$\det(I + Q(s)) = \prod_{\gamma \in \mathcal{P}} \left(1 - (-1)^{|\gamma|} q_s(\gamma)\right), \quad (2)$$

absolutely convergent, with the acyclic-composability and SCC factorizations as corollaries, and a truncation-by-length bound $|\text{tail}_{>p}| \leq m \bar{q}^{p+1} / [(p+1)(1-\bar{q})]$, $\bar{q} = \rho(|Q|)$.

3 The portfolio spectral game

3.1 Set function and restriction consistency

Fix a study frequency s (or a band; all statements are pointwise in s and can be integrated). For $S \subseteq \mathcal{A}$ let K_S be the action Green matrix of the sub-portfolio S , and define the *portfolio spectral game*

$$\Phi(S) = \log \det(I + K_S(s)), \quad \Phi(\emptyset) = 0. \quad (3)$$

$\Phi(S)$ is the exact log spectral shift (Krein perturbation determinant) produced by executing exactly the projects in S against the common base grid.

Lemma 1 (Restriction consistency). $K_S = K_{\mathcal{A}}[S, S]$ and $Q_S = Q_{\mathcal{A}}[S, S]$: the interaction data of every sub-portfolio are principal submatrices of the full-cluster data. Moreover $\rho(|Q_S|) \leq \rho(|Q_{\mathcal{A}}|)$.

Proof. $K_{\mathcal{A}} = CV^*G_0U$ with one column of U , row of V^* , and diagonal entry of C per action; deleting the actions outside S deletes the corresponding rows and columns and changes nothing else, since G_0 is the *unperturbed* grid. The diagonal normalization in Q is per-action, so it commutes with restriction. The radius inequality is the standard principal-submatrix monotonicity of the Perron root of a nonnegative matrix [20]. $\square$

One evaluation of G_0 therefore prices every coalition: the game (3), which naively requires 2^m grid studies, is encoded in a single $m \times m$ matrix.

3.2 Dividends are prime-cycle bundles

For a set function v with $v(\emptyset) = 0$, the Harsanyi dividends (Möbius transform) are $\mu_v(R) = \sum_{R' \subseteq R} (-1)^{|R|-|R'|} v(R')$, and $v(S) = \sum_{R \subseteq S} \mu_v(R)$ [6, 7]. Dividends are the intrinsic synergy of each coalition, purged of all sub-coalition effects; they are the canonical basis of cooperative game theory.

Theorem 1 (Dividend–cycle correspondence). Assume $\rho(|Q_{\mathcal{A}}(s)|) < 1$ and $1 + K_{aa}(s) \neq 0$ for all a . Then the Harsanyi dividends of the portfolio spectral game are

$$\mu_{\Phi}(\{a\}) = \log(1 + K_{aa}(s)), \quad (4)$$

$$\mu_{\Phi}(R) = \sum_{\substack{\gamma \in \mathcal{P} \\ \text{supp}(\gamma) = R}} \log(1 - (-1)^{|\gamma|} q_s(\gamma)), \quad |R| \geq 2, \quad (5)$$

with the inner sum absolutely convergent. In particular, for every pair $\{a, b\}$ the dividend is a single closed-form factor,

$$\mu_{\Phi}(\{a, b\}) = \log(1 - q_{ab}q_{ba}), \quad (6)$$

because the only prime cycle supported on two actions is the 2-cycle.

Proof. By Lemma 1 and (2) applied to Q_S ,

$$\Phi(S) = \sum_{a \in S} \log(1 + K_{aa}) + \sum_{\substack{\gamma \in \mathcal{P} \\ \text{supp}(\gamma) \subseteq S}} \log(1 - (-1)^{|\gamma|} q(\gamma)),$$

since the prime cycles of the restricted digraph are exactly the prime cycles of the full digraph with support inside S , with unchanged weights. Grouping the cycle sum by exact support exhibits $\Phi(S) = \sum_{R \subseteq S} m(R)$ with m given by the right-hand sides above; by uniqueness of the Möbius representation on the subset lattice, $\mu_{\Phi} = m$. Absolute convergence for each R follows from $\sum_{|\gamma|=\ell} |q(\gamma)| \leq \text{tr}(|Q|^{\ell})/\ell \leq m\bar{q}^{\ell}/\ell$. For (6): on two nodes every closed walk alternates and is a power of the 2-cycle, whose only primitive representative is the 2-cycle itself (Appendix A). $\square$

Remark 1. Equation (6) gives the empirically measured pair Möbius interactions of [2] an exact identity: the pair kernel $K_{ef} = |\beta_e \beta_f r_{ef} r_{fe}|$ that detected the strongest 5% of non-additive pairs with AUC ≈ 0.96 is precisely $|q_{ef} q_{fe}|$ in a dominant-mode approximation—the argument of the closed-form dividend. Triple dividends start at the two oriented 3-cycles and carry an infinite, certified-truncatable tail (support- $\{a, b, c\}$ primes of length ≥ 4 such as $a \rightarrow b \rightarrow a \rightarrow c \rightarrow a$).

3.3 Closed-form Shapley attribution

The Shapley value distributes each dividend equally among its coalition [5, 6]: $\varphi_a = \sum_{R \ni a} \mu_v(R)/|R|$.

Theorem 2 (Cycle-split Shapley formula). Under the hypotheses of Theorem 1, the Shapley value of action a in the portfolio spectral game is

$$\varphi_a = \log(1 + K_{aa}) + \sum_{\substack{\gamma \in \mathcal{P} \\ a \in \text{supp}(\gamma)}} \frac{\log(1 - (-1)^{|\gamma|} q(\gamma))}{|\text{supp}(\gamma)|} \quad (7)$$

with $\sum_a \varphi_a = \Phi(\mathcal{A})$ (efficiency). Truncating the cycle sum at length p incurs error at most $m\bar{q}^{p+1}/[2(p+1)(1-\bar{q})]$ per action, and the truncated values are computed by enumerating Lyndon words over the action alphabet up to length p , in $O(m^p)$ arithmetic operations. By contrast, computing Shapley values of general succinctly represented games is #P-hard [8]; the portfolio spectral game is tractable because its dividends are supported on cycle bundles.

Interpretation: each project owes (or is owed) its isolated spectral effect, plus an equal share of every closed feedback loop it participates in. Three standard axioms become engineering statements. *Null player*: a project that closes no cycle with others—in particular any project outside every SCC feedback core—receives exactly its isolated term; this is the but-for causation standard, derived. *Symmetry*: projects entering the game identically are

charged identically. *Additivity*: attribution over disjoint frequency bands or independent margin functionals adds.

3.4 Cluster-composition independence

Theorem 3. *Fix the base grid T_0 (the common study snapshot). For any two clusters $\mathcal{A}' \supseteq R$ and $\mathcal{A}'' \supseteq R$ containing coalition R , the dividend $\mu_\Phi(R)$ computed within $\mathcal{A}'$ equals that computed within $\mathcal{A}''$: coalition synergies are local to their members. Consequently, adding or withdrawing unrelated projects from the queue cannot change any coalition’s assessed responsibility, and can change a project’s Shapley value only through cycles that the entering/leaving projects themselves participate in.*

Proof. Immediate from Theorem 1: $\mu_\Phi(R)$ depends only on entries q_{ab} with $a, b \in R$, which by Lemma 1 depend only on G_0 and the members of R . $\square$

This property is not decorative. Pro-rata rules (per MW, per electrical distance) recompute every share when the cluster changes, creating well-documented incentives to time entries and withdrawals; under (7) a project’s exposure to a given feedback core is invariant to queue churn outside that core. The caveat is explicit: the statement holds per study snapshot with a common base; once projects are actually energized, the base G_0 is updated and the game is re-posed.

4 Whose mode is it? Integer attribution

Allocation of a scalar margin shift is often not the contested question; the contested question after an event is *which projects created the oscillatory mode*. The Euler product answers it in integers.

Theorem 4 (Mode-count attribution). *Let $\partial\Omega$ be a rectifiable Jordan contour in the s -plane on which $T_0(s)$ is invertible, $1 + K_{aa}(s) \neq 0$, each Euler factor is nonvanishing (generic), and $\rho(|Q(s)|) \leq 1 - \delta$ for some $\delta > 0$. Then the net number of closed-loop modes created minus destroyed by the portfolio inside Ω satisfies*

$$N_{\mathcal{A}}(\Omega) - N_0(\Omega) = \sum_a \text{wind}_{\partial\Omega}(1 + K_{aa}) + \sum_{\gamma \in \mathcal{P}} w_\gamma, \quad (8)$$

where $w_\gamma \in \mathbb{Z}$ is the winding number of the Euler factor of γ along $\partial\Omega$, and all but finitely many w_γ vanish. The finitely many prime cycles with $w_\gamma \neq 0$ are explicitly identifiable and constitute the exact causal inventory of the interaction-created modes in Ω .

Proof sketch. By the argument principle, $N_{\mathcal{A}} - N_0 = \text{wind}_{\partial\Omega} \det(I + K)$, and $\det(I + K) = \prod_a (1 + K_{aa}) \det(I + Q)$. On the contour, $|q_s(\gamma)| \leq m(1 - \delta)^{|\gamma|}$, so the product (2) converges uniformly on $\partial\Omega$ and its logarithm converges uniformly; the winding of the product equals the sum of

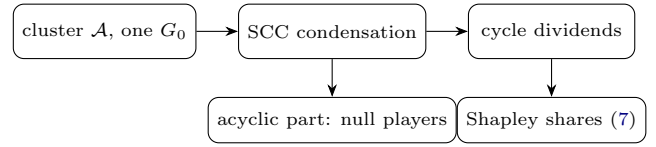


Figure 1: Allocation pipeline. One linearization prices all 2^m coalitions; SCC condensation separates null players from feedback cores; dividends and shares follow in closed/truncated form.

factor windings. Each factor is continuous and nonvanishing on the closed curve, so its winding is an integer; for $|\gamma|$ beyond a computable finite length the factor lies in a disc of radius < 1 around 1 and its winding is zero. Details in Appendix A. $\square$

The regulatory reading: on a certified contour surrounding the observed mode, “whose oscillation” reduces to listing the prime cycles with nonzero winding—an auditable, integer-valued computation, replacing negotiation. Where certification fails ($\rho(|Q|) \geq 1$ on the contour), the theorem is silent, and that silence is itself informative: it flags exactly the frequency bands where only a joint study can adjudicate.

5 Cluster decomposition and the allocation rule

5.1 Study-group decomposition

By the SCC corollary of (2), $\det(I + Q)$ factors over the strongly connected components of the action digraph; combined with Theorem 1, every nonzero dividend with $|R| \geq 2$ lives inside a single SCC. The cluster study therefore decomposes exactly: the acyclic majority of projects is screened independently (null players for the interaction), and joint dynamic studies are required only within feedback cores—collapsing a combinatorial 2^m study into $\sum_c O(2^{|c|})$ or, with certified truncation, $O(m^p)$ evaluations of one matrix.

5.2 The allocation rule and its comparison class

Rule: allocate the cluster’s dynamic-security externality (margin shift, or the mitigation cost occasioned by it) according to the Shapley shares (7); attribute created modes by Theorem 4. *Properties*: efficiency (shares sum to the whole), symmetry, additivity across bands and functionals, null player = but-for causation, cluster-composition independence (Theorem 3), and exact computability. Pro-rata per MW satisfies efficiency only; it charges projects that close no cycle, exempts none, and shifts with queue churn. Sequential (queue-order) allocation is order-dependent by construction; the Shapley value is its order average, computed here without averaging.

Table 1: Flagship destructive portfolio [2]: margin game $F(S) = \zeta_{\min}(S)$, dividends, and Shapley shares.

$F(\emptyset)$	0.045218
$F(e)$ line 26–29	0.045872 (+1.45%)
$F(c)$ retuning 5.1 $\rightarrow$ 4.8 rad/s	0.047657 (+5.39%)
$F(ec)$ joint	0.040352 (−10.76%)
$\mu(\{e\}), \mu(\{c\})$	+0.000654, +0.002439
$\mu(\{e, c\})$ (mixed cycle)	−0.007959
$\varphi_e = \mu(e) + \frac{1}{2}\mu(ec)$	−0.003325 (68.3%)
$\varphi_c = \mu(c) + \frac{1}{2}\mu(ec)$	−0.001541 (31.7%)
$\varphi_e + \varphi_c = F(ec) - F(\emptyset)$	−0.004866 ✓

5.3 Worked flagship example

Table 1 applies the rule to the two-project destructive portfolio. Both projects are individually beneficial, yet each ends with a *negative* net share: the mixed line–controller prime cycle (dividend −0.007959, the hybridization-driven pair of [2], where the indicator rises from 0.381 to 1.121 jointly) dominates both isolated benefits. The line bears 68.3% of the net damage and the retuning 31.7%—not because the line is “worse,” but because its isolated benefit offsets less of its half of the cycle. A third project outside the cycle would, by the null-player axiom, owe nothing. In the exact- Φ game the pair dividend is the closed form (6); here the margin functional $\zeta_{\min}$ is used directly (Shapley applies to any set function), and the bridge between the two functionals is quantified below.

6 Cycle surgery: dissolving the core

Attribution invites prescription: rather than pricing a destructive cycle, remove it at minimum cost.

Proposition 1 (Monotone, targetable radius). *Let $\bar{q}(\theta) = \rho(|Q(\theta)|)$ depend on design parameters θ (controller gains, filter settings, action sizings) through the entries $|q_{ab}|$. Then: (i) $\bar{q}$ is nondecreasing in each $|q_{ab}|$, so any surgery that only shrinks cycle-edge magnitudes certifiably shrinks the radius; (ii) if $|Q|$ is irreducible on a core with left/right Perron vectors u, v ($u^\top v = 1$),*

$$\frac{\partial \bar{q}}{\partial |q_{ab}|} = u_a v_b, \quad (9)$$

so the cost-optimal first-order surgery shrinks the edge maximizing $u_a v_b / \text{cost}_{ab}$; (iii) rendering the action digraph acyclic by edge suppression is a minimum feedback arc set problem, NP-hard in general [21] but trivially solvable by integer programming at cluster-core sizes.

Both facts are classical Perron–Frobenius theory [20]; the contribution is the design loop they enable: identify the destructive core (Section 5), read the Perron-sensitive

edges, and buy the cheapest parameter change that either drives $\bar{q}$ below $1 - \varepsilon$ on the band or breaks the cycle outright. The IEEE 39-bus stress case already exhibits the pattern empirically: after a connectivity-improving line destabilized a coupled GFL mode, retuning a *single* attributed PLL ($K_{p,30}^* = 2.108$) restored the margin without undoing the upgrade [3]. In the present language that retuning is a rank-one surgery on the dominant Perron edges of the mixed core—diagnosis and repair now share one calculus.

7 Evidence, fidelity of the proxy, and calibration

No new simulations are reported; the mechanism-level IEEE 39-bus campaigns [2, 4] are read through the new theorems.

Exactness checks. The determinant identities underlying (3) were verified to relative error 5.8×10^{-15} ; the acyclic/SCC factorizations to 2.22×10^{-16} over 500 random tests; the truncation bound over 4,800 truncations with zero violations (max error-to-bound ratio 0.358).

Fidelity of the spectral game for margin allocation. The exact attribution lives on $\Phi = \log \det(I + K)$; planners price $\zeta_{\min}$. The campaigns quantify the bridge: the determinant observable correlates with damping non-additivity at Spearman 0.673 (AUC 0.897, 17,664 line/controller pairs) and 0.747 (AUC 0.894, 23,000 GFM-like pairs); the pair and triple cycle kernels track the margin Möbius interactions at Spearman 0.772 (AUC ≈ 0.96) and 0.874. Attribution on Φ is exact; attribution on $\zeta_{\min}$ is either computed directly (as in Table 1, at $2^{|c|}$ cost inside small cores) or proxied by the cycle formula with the above measured fidelity. Both routes are legitimate; the paper keeps them distinct.

Scaling of exposure under decarbonization. Under co-retirement of inertia and damping $M, D \mapsto \alpha(M, D)$, every prime-cycle weight scales as $\alpha^{-|\gamma|/2}$ (fitted slopes −0.989, −1.469 against predicted −1, −3/2, $R^2 > 0.99989$) [2]. Dividends and Shapley exposures therefore grow in strict order of cycle length as synchronous machines retire: the allocation problem provably intensifies along the energy transition, and reverses if absolute damping is preserved—a quantitative argument for procuring damping as a product.

Limitations. (i) All grid evidence is mechanism-level (reduced SG/GFL/GFM surrogates on re-solved AC operating points); manufacturer-grade EMT replication is the external validation gate. (ii) The equal within-coalition split is the Shapley axiom package; a visit-multiplicity split is an alternative and would require its own axiomatization. (iii) Monetary allocation of a physical externality is a policy layer on top of the responsibility shares; the theorems allocate responsibility, not tariffs. (iv) Theorem 4 requires certification on the contour; bands where $\rho(|Q|) \geq 1$ demand joint study by construction. (v) The-

orem 3 holds per study snapshot with a common base.

Falsification tests for a full-order campaign: (a) do closed-form pair dividends (6) track full-model pairwise non-additivity within the certified region; (b) does the winding inventory of Theorem 4 identify the devices found by biorthogonal participation in an engineered destructive case; (c) does Perron-targeted surgery repair a destructive core at lower parameter cost than exhaustive retuning; (d) do Shapley exposures grow with the predicted α -exponents under staged machine retirement. Failure narrows the industrial claims without touching the exact identities.

8 Regulatory interpretation

Cluster-study reforms [12] already force joint evaluation; what they lack is a causal division of the jointly caused. The rule proposed here slots into that gap with properties a regulator can state: every dollar of interaction-occasioned mitigation is traceable to a named feedback loop among named projects; projects outside every loop are certified exempt; the assessment of a loop cannot be moved by queue churn outside it; and after an event, the created mode itself is attributed in integers on a certified contour. None of this replaces EMT validation of final designs; it replaces the negotiation that today substitutes for attribution. The same calculus prices the alternative to paying: the cheapest cycle surgery is a computable quantity, so “mitigate” versus “allocate” becomes a comparable pair of numbers.

9 Conclusion

A cluster of grid projects is a cooperative game whose characteristic function is a perturbation determinant, and that game is exactly solvable: its Harsanyi dividends are prime intervention cycles, its Shapley values are closed-form cycle splits, its null players are the certified non-participants in any feedback core, its created modes are integer windings of finitely many identifiable cycles, and its cheapest repair is a Perron-targeted surgery. The question that today ends in negotiation—*who pays for the oscillation?*—is, within the certified region of the action zeta function, a theorem.

A Proof details

A.1 Pair dividends (Eq. (6))

On support $\{a, b\}$ every closed walk alternates $a, b, \dots$ and is the k -fold traversal of the 2-cycle (ab) ; the unique prime with that support is (ab) with weight $q_{ab}q_{ba}$ and sign $(-1)^2 = +1$, giving $\mu(\{a, b\}) = \log(1 - q_{ab}q_{ba})$. Consistently, $\det(I + Q_{\{a, b\}}) = 1 - q_{ab}q_{ba}$ exactly.

A.2 Theorem 4

Write $\det(I + K) = \prod_a (1 + K_{aa}) \cdot P(s)$ with $P(s) = \prod_\gamma f_\gamma(s)$, $f_\gamma = 1 - (-1)^{|\gamma|} q_s(\gamma)$. On $\partial\Omega$, $|q_s(\gamma)| \leq (|Q|^{|\gamma|})_{aa} \leq \text{tr}(|Q|^{|\gamma|}) \leq m(1 - \delta)^{|\gamma|}$, so $\sum_\gamma \sup_{\partial\Omega} |1 - f_\gamma| \leq \sum_{\ell \geq 2} m(1 - \delta)^\ell < \infty$: the product and $\log P = \sum \log f_\gamma$ converge uniformly on the compact contour (finitely many short factors handled directly under the genericity hypothesis $f_\gamma \neq 0$ on $\partial\Omega$). Uniform convergence of the logarithms gives $\text{wind}(P) = \sum_\gamma \text{wind}(f_\gamma)$; each $\text{wind}(f_\gamma)$ is an integer since f_γ is continuous and non-vanishing on a closed curve; whenever $\sup_{\partial\Omega} |1 - f_\gamma| < 1$ the image omits 0’s ray neighborhood and $\text{wind}(f_\gamma) = 0$, which holds for all $|\gamma| > \log m / \log \frac{1}{1-\delta}$. The argument principle applied to the meromorphic $\det(I + K) = \det T_A / \det T_0$ identifies $\text{wind} \det(I + K)$ with $N_A - N_0$ inside Ω .

A.3 Shapley truncation error

Each omitted cycle contributes $|\log f_\gamma| / |\text{supp } \gamma| \leq |\log f_\gamma| / 2$ to at most $|\text{supp } \gamma|$ actions; summing the tail as in the truncation bound of [1] gives the per-action bound in Theorem 2.

B Dividend calculus for three actions

For $\mathcal{A} = \{a, b, c\}$: pair dividends are the three closed forms (6); the triple dividend is $\mu(\{a, b, c\}) = \log(1 + q_{ab}q_{bc}q_{ca}) + \log(1 + q_{ac}q_{cb}q_{ba}) + \sum_{|\gamma| \geq 4, \text{supp}=\{a, b, c\}} \log f_\gamma$, with tail certified by the length bound. Shapley: $\varphi_a = \mu(a) + \frac{1}{2}[\mu(ab) + \mu(ac)] + \frac{1}{3}\mu(abc)$, etc. The interaction-aware triple of [2]—three top-ranked lines jointly -6.87% versus an interaction-aware triple $+19.59\%$ —is the statement that portfolio optimization should search over dividends, not marginal ranks.

References

- [1] J. L. Mayorga Taborda, “Prime intervention cycles: an Euler-product zeta certificate for corrective-action portfolios in inverter-dominated power grids,” companion manuscript, Aug. 2026.
- [2] J. L. Mayorga Taborda, “Controller-mediated intervention cycles in inverter-dominated power grids,” manuscript, Aug. 2026.
- [3] J. L. Mayorga Taborda, “Gauge-invariant scalarizability of multi-inverter power networks,” preprint draft v4, Aug. 2026.
- [4] J. L. Mayorga Taborda, “Two-gate decision trust in inverter-dominated power grids,” unified manuscript, Aug. 2026.

- [5] L. S. Shapley, “A value for n -person games,” in *Contributions to the Theory of Games II*, Princeton Univ. Press, 1953, pp. 307–317.
- [6] J. C. Harsanyi, “A bargaining model for the cooperative n -person game,” in *Contributions to the Theory of Games IV*, Princeton Univ. Press, 1959, pp. 325–355.
- [7] M. Grabisch and M. Roubens, “An axiomatic approach to the concept of interaction among players in cooperative games,” *Int. J. Game Theory*, vol. 28, pp. 547–565, 1999.
- [8] X. Deng and C. H. Papadimitriou, “On the complexity of cooperative solution concepts,” *Math. Oper. Res.*, vol. 19, no. 2, pp. 257–266, 1994.
- [9] S. M. Lundberg and S.-I. Lee, “A unified approach to interpreting model predictions,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2017.
- [10] A. Churkin, J. Bialek, D. Pozo, E. Sauma, and N. Korgin, “Review of cooperative game theory applications in power system expansion planning,” *Renew. Sustain. Energy Rev.*, vol. 145, 2021.
- [11] J. Bialek, “Tracing the flow of electricity,” *IEE Proc.—Gener. Transm. Distrib.*, vol. 143, no. 4, pp. 313–320, 1996.
- [12] Federal Energy Regulatory Commission, *Improvements to Generator Interconnection Procedures and Agreements*, Order No. 2023, Docket RM22-14, 2023.
- [13] NERC, *Odessa Disturbance: Texas Events, May 9, 2021 and June 26, 2021 – Joint NERC and Texas RE Staff Report*, 2021; and *2022 Odessa Disturbance Report*, 2022.
- [14] National Grid ESO, *Technical Report on the events of 9 August 2019*, Sept. 2019.
- [15] R. Bowen and O. Lanford, “Zeta functions of restrictions of the shift transformation,” *Proc. Symp. Pure Math.*, vol. 14, pp. 43–49, 1970.
- [16] H. M. Stark and A. A. Terras, “Zeta functions of finite graphs and coverings,” *Adv. Math.*, vol. 121, pp. 124–165, 1996.
- [17] D. Foata and D. Zeilberger, “A combinatorial proof of Bass’s evaluations of the Ihara–Selberg zeta function for graphs,” *Trans. Amer. Math. Soc.*, vol. 351, pp. 2257–2274, 1999.
- [18] A. Terras, *Zeta Functions of Graphs: A Stroll through the Garden*. Cambridge Univ. Press, 2010.
- [19] Y. Watanabe and K. Fukumizu, “Graph zeta function in the Bethe free energy and loopy belief propagation,” *NeurIPS*, 2009; arXiv:1002.3307.
- [20] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed. Cambridge Univ. Press, 2013.
- [21] R. M. Karp, “Reducibility among combinatorial problems,” in *Complexity of Computer Computations*, Plenum, 1972, pp. 85–103.
- [22] W. Du, X. Wang, and H. Wang, “Sub-synchronous interactions caused by the PLL in the grid-connected PMSG for wind power generation,” *Int. J. Electr. Power Energy Syst.*, vol. 98, pp. 331–341, 2018.
- [23] J. Kook and J.-W. Park, “An interaction sensitivity framework in perturbation space for the converter-driven stability of power systems,” arXiv:2607.25334, 2026.
- [24] T. Joswig-Jones, S. Dong, J. Tan, and B. Zhang, “Sensitivity-based system strength assessment,” arXiv:2607.28764, 2026.